SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION

Course Code:	CSA16	**Course Title:**	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 2 – Technical Presentation
Weightage:	10% of CIA	**Total Marks:**	100 (2 Question × 100 Marks)
CO1 Statement:	*Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2*		
Student Name:	TEJASWINI.M	**Reg. No.:**	192411079
Section / Batch:	BECSE (2024)	**Date:**	15-07-26

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

Question Type	Focus Area	Questions	Marks
Technical Presentation	Decision support system, Operational versus Decision support System, Architecture of data warehouse, ETL, OLAP and data cubes, Dimensional data modelling -star snowflake schemas. Query Processing, Open-source Business Intelligence Tools. Develop an application to perform OLAP operations - Introduction to KNIME tool for reporting.	Q1, Q2	100
		GRAND TOTAL:	**100 Marks**

SECTION A – Technical Presentation

Topic 1: Decision Support Systems (DSS)

🎯 Focus Areas:

- Definition and need of DSS
- Components (Data, Model, UI)
- Types of DSS
- Real-world applications (banking, healthcare)
- Advantages & limitations
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Topic 2: Operational Systems vs DSS

🎯 Focus Areas:

OLTP vs OLAP concepts
Key differences (data, users, purpose)
Performance comparison
Use-case scenarios
Integration between systems

Topic 3: Architecture of Data Warehouse
🎯 Focus Areas:
- Data warehouse definition
- Three-tier architecture
- ETL process
- Data marts & metadata
- Benefits in decision-making

Topic 4: OLAP and Data Cubes
🎯 Focus Areas:
- OLAP concept
- Data cube structure
- OLAP operations (slice, dice, roll-up, drill-down)
- Applications in analytics

Topic 5: Dimensional Data Modeling
🎯 Focus Areas:
- Fact and dimension tables
- Star schema vs snowflake schema
- Schema design process
- Real-world example

Topic 6: Query Processing & BI Tools
🎯 Focus Areas:
- Query processing stages
- Query optimization
- BI tools overview
- Reporting and visualization
- Examples of open-source BI tools

Topic 7: OLAP Application using KNIME
🎯 Focus Areas:
- Introduction to KNIME
- Workflow components (nodes, connections)
- Performing OLAP operations
- Sample application/demo
- Report generation

Code of Conduct

I TEJASWINI. M (192411079) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M. Tejaswini

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Marks Evaluation:

Question No.	Technical Content Accuracy (30%)	Problem Identification (25%)	Use of Tools (20%)	Communication (15%)	Professionalism (10%)	Total Marks (100)
Q1						
Q2						
					Overall Total (100)	

Faculty In-charge

Signature: ____________

Date: ____________

Course Coordinator

Signature: ____________

Date: ____________

Head of Department

Signature: ____________

Date: ____________

ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION

CSA1610 - DATA WAREHOUSING AND DATA MINING

SECTION A – Technical Presentation

Presentation Summary

- Decision Support System (DSS) helps managers make accurate and faster decisions using data analysis.
- DSS consists of three main components: Data, Model, and User Interface.
- Different types of DSS support various business needs, such as data analysis, expert recommendations, and team decision-making.
- DSS is widely used in banking, healthcare, retail, manufacturing, and other industries to improve decision-making.
- DSS improves productivity and decision quality, but it depends on accurate data and supports rather than replaces human judgment.

SIMATS ENGINEERING SIMATS

Decision Support Systems (DSS)

CSA1610 - DATA WAREHOUSING AND DATA MINING

PRESENTED BY:

TEJASWINI M (192411079)

PROBLEM & DEFINITION

Problem

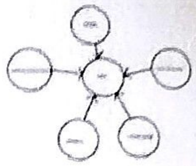
- Huge amount of business data
- Difficult decision-making
- Time pressure
- Risk of human errors

Definition

A computer-based system that helps managers make better decisions using data and analytical models.

Objective

- Better decisions
- Faster decisions
- Accurate decisions



COMPONENTS OF DSS

Data

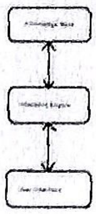
- Database
- Data Warehouse
- External Sources

Model

- Mathematical Models
- Statistical Analysis
- Algorithms

User Interface

- Reports
- Charts
- Dashboards



TYPES OF DSS

[illegible]	[illegible]	[illegible]
Data-driven	[illegible]	[illegible]
Model-driven	[illegible]	[illegible]
Communication-driven and group	[illegible]	[illegible]
Knowledge-driven	[illegible]	[illegible]
Document-driven	[illegible]	[illegible]

ADVANTAGES & LIMITATIONS

Advantages

✓ Better decision accuracy
✓ Saves time
✓ Reduces human errors
✓ Quick data analysis
✓ Supports strategic planning

Limitations

✗ High implementation cost
✗ Depends on data quality
✗ Needs trained users
✗ Cannot replace human judgment

CONCLUSION

Conclusion

✓ Converts data into useful information
✓ Supports better decision-making
✓ Improves productivity
✓ Helps business growth

Future Scope

Artificial Intelligence (AI)
Machine Learning (ML)
Big Data
Cloud Computing